BUSINESS & FUNCTIONAL REQUIREMENTS

RetailCo Analytics Platform

Data Modeling Initiative - Customer, Order & Product Domains

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1 Executive Summary

RetailCo is a multi-channel retailer operating across web, mobile, in-store, and partner channels. The business has identified that fragmented customer data across three legacy operational systems is limiting its ability to understand customer purchase behavior, measure marketing return on investment, and support data-driven decision-making at scale.

This document defines the business and functional requirements for a data modeling initiative that will establish the foundational data structures for the RetailCo Analytics Platform. The initiative covers three core business domains - Customer, Order, and Product - and will produce data models at conceptual, logical (third normal form), and physical (star schema for Snowflake) layers.

The primary business objective is to enable unified analysis of customer purchase behavior to improve marketing ROI by 15% within 12 months of platform deployment. Secondary objectives include cloud migration readiness, self-service analytics enablement, and establishment of a governed data foundation that can support future data science and AI workloads.

2 Business Context & Strategic Drivers

2.1 Organization Overview

RetailCo sells products across four channels: a transactional website, a mobile application, physical retail stores, and a partner network. Customers can purchase through any combination of channels, and the business tracks customer journeys and purchase histories independently in each system. The absence of a unified customer record means that a customer who shops both online and in-store is treated as two separate customers in current reporting.

2.2 Strategic Drivers

- Customer intelligence: The marketing team cannot determine true customer lifetime value, channel preference, or segment transitions because customer identity is not unified across systems.
- Cloud migration: The existing on-premises data warehouse and ETL infrastructure is being decommissioned. All analytical workloads must move to a cloud platform (Snowflake on Azure) by end of the financial year.
- Self-service analytics: Business analysts require access to curated, governed data they can explore without depending on the data engineering team for every query.
- Data governance: Regulatory and audit requirements mandate that personally identifiable information (PII) be classified, tracked, and masked in non-production environments. Currently no formal data classification exists.
- AI and ML readiness: The data science team is blocked from building customer propensity and churn models because training datasets cannot be reliably assembled from the fragmented source systems.

3 Business Objectives & Success Criteria

3.1 Primary Objective

Primary Goal	Understand customer purchase behavior across all channels to improve marketing ROI by 15% within 12 months of platform deployment, through unified customer identity, accurate lifetime value calculation, and channel attribution.

3.2 Supporting Objectives

- Establish a single, authoritative customer record that persists identity across channels and captures changes over time.
- Enable accurate calculation of order-level and line-item-level revenue, discount, and margin metrics.
- Support product performance analysis through a three-level category hierarchy and supplier attribution.
- Provide a governed, cloud-native data foundation deployable on Snowflake that can serve BI, self-service analytics, and future ML workloads.
- Classify and protect PII from day one, with column-level masking enforced in all non-production environments.

3.3 Success Criteria

ID	Priority	Requirement
SC-01	Must	A unified customer record can be queried that returns accurate lifetime value regardless of the channels through which the customer has purchased.
SC-02	Must	Order revenue, discount, and net revenue figures in the analytics platform reconcile to source system totals within a 0.01% tolerance.
SC-03	Must	PII attributes are classified in the data model and masked in all non-production environments before first data load.
SC-04	Must	The data model forward-engineers to valid Snowflake DDL without manual modification.
SC-05	Should	Customer segment transitions are captured and queryable historically, enabling cohort analysis by segment at any point in time.
SC-06	Should	Product category hierarchy is navigable at three levels (Root, Mid, Leaf) in all BI tools connected to the platform.
SC-07	Could	Gross margin per product per order line is calculable for finance roles who have access to standard cost data.

4 Business Domain Narratives

The following narratives describe how RetailCo operates within each domain in business language. These narratives are the primary input to the data modeling process. The data modeler should derive entities, attributes, relationships, and constraints from these descriptions without requiring additional clarification.

4.1 Customer Domain

A customer is any person or organization that has registered with RetailCo or completed a purchase. Each customer has a unique identity across all channels, identified by a system-assigned identifier. Customers provide a primary email address at registration, which is unique across all customers and serves as the natural contact key. First name and last name are captured at registration.

Each customer is assigned to a segment - Gold, Silver, Bronze, or New - based on their cumulative lifetime value (the sum of all net revenue attributed to that customer across all orders). Segment thresholds are defined by the marketing team and may change over time; when a customer crosses a threshold, their segment must be updated. The system must retain a history of segment changes so that cohort analyses can be performed at any historical point in time.

How a customer was first acquired is recorded as an acquisition channel (Web, Store, Referral, or Partner). This attribute does not change once set. A customer may have one or more physical addresses on file, each classified as a billing address, a shipping address, or both. Exactly one address per customer is designated as the primary billing address. Addresses include street, city, postal code, and country.

Customers may be deactivated rather than deleted. An inactive customer retains all historical order data. The date and time of first registration and most recent profile update must be recorded for audit purposes.

4.2 Order Domain

An order is a purchase transaction initiated by a customer through one of the four channels (Web, Mobile, In-Store, or Partner). Each order is uniquely identified and linked to exactly one customer. An order passes through a defined lifecycle: Pending, Confirmed, Shipped, Delivered, and Cancelled. The timestamp of order creation is recorded in UTC.

An order may optionally have a single promotion applied to it. A promotion provides a discount that may be expressed as a percentage of the order value, a fixed monetary amount, or a buy-one-get-one (BOGO) arrangement. The gross order value (the sum of line item extended prices before any order-level discount) and the discount amount are both recorded on the order. Net revenue is derived as gross value minus discount amount.

Each order contains one or more line items. A line item records the product purchased, the quantity ordered (which must be greater than zero), the unit price at the time of the order (which may differ from the product's current list price), and any line-level discount expressed as a fraction between 0 and 1.

Extended price for a line item is derived as quantity multiplied by unit price multiplied by one minus the discount fraction. Line-level discounts are distinct from the order-level promotion discount.

An order must contain at least one line item. Deleting an order must cascade to its line items. A product may appear on many line items across many orders, but each line item references exactly one product.

4.3 Product Domain

A product is a sellable item in the RetailCo catalogue. Each product is identified by a system-assigned UUID and carries a stock-keeping unit (SKU) code that is unique across all products. Product name and current list price are maintained by the merchandising team. Products may be deactivated when discontinued; deactivated products must be retained so that historical order line items remain navigable.

Each product belongs to exactly one category, which must be a leaf-level category (the lowest level in the hierarchy). Categories are organized in a self-referencing hierarchy with up to three levels: Root (e.g. Electronics), Mid (e.g. Computing), and Leaf (e.g. Laptops). A root category has no parent. Each category has a name and a level indicator. Products may only be assigned to level-3 (Leaf) categories.

Each product is sourced from a primary supplier. A supplier record captures the supplier name, country of operation, standard lead time in calendar days, and an optional procurement contact email. A product may only have one primary supplier at a time, though this may change; the current supplier relationship is what is tracked (no supplier history is required).

Cost price (the per-unit cost from the supplier) is optionally recorded on the product and is a financially sensitive attribute accessible only to finance roles. List price is the standard selling price; the actual price charged on a given line item is captured as unit price on the line item and may differ from the current list price.

5 Functional Requirements

5.1 Customer Management

ID	Priority	Requirement
FR-C01	Must	The system must store a unique, system-assigned identifier for each customer that persists across all channels and is used as the primary key in all downstream systems.
FR-C02	Must	Email address must be stored, validated as unique across all customers, and flagged as PII requiring masking in non-production environments.
FR-C03	Must	First name and last name must be stored and flagged as PII.
FR-C04	Must	Each customer must be assigned to exactly one segment (Gold, Silver, Bronze, or New) at any point in time.

ID	Priority	Requirement
FR-C05	Must	The acquisition channel (Web, Store, Referral, or Partner) must be recorded once at registration and must not be updated thereafter.
FR-C06	Must	Customer lifetime value must be stored as a numeric field and updated to reflect cumulative net revenue.
FR-C07	Must	Customers must be soft-deletable via an IsActive flag; physical deletion of customer records is not permitted.
FR-C08	Must	UTC timestamps for record creation and last modification must be maintained for audit purposes.
FR-C09	Must	Each customer may have one or more addresses. Each address must be typed as Billing, Shipping, or Both. Exactly one address per customer must be designated as primary.
FR-C10	Must	Address must capture street, city, postal code, and country.
FR-C11	Should	Customer segment history must be retained so that the segment assignment of any customer at any past date can be determined. Segment changes must be recorded with effective-from and effective-to dates.

5.2 Order Management

ID	Priority	Requirement
FR-O01	Must	Each order must be uniquely identified and linked to exactly one customer.
FR-O02	Must	The channel through which an order was placed (Web, Mobile, In-Store, or Partner) must be recorded and must not change after order creation.
FR-O03	Must	Order status must be one of: Pending, Confirmed, Shipped, Delivered, or Cancelled. Status transitions must be recorded; the current status is what is stored.
FR-O04	Must	The UTC timestamp of order creation must be recorded.
FR-O05	Must	Gross order value (sum of line item extended prices before order-level discount) must be stored. Discount amount must be stored separately. Net revenue is derived and need not be stored on the order record.
FR-O06	Must	An order may reference at most one promotion. The promotion reference is optional; an order without a promotion has a null promotion reference.
FR-O07	Must	Each order must contain at least one line item.
FR-O08	Must	Each line item must record the product, quantity (greater than zero), unit price at time of order, and a line-level discount fraction (0 to 1, defaulting to 0).

ID	Priority	Requirement
FR-O09	Must	Extended price per line item is derived as Quantity x UnitPrice x (1 - DiscountPct) and must be available for analytical queries. It may be stored as a generated column.
FR-O10	Must	Deletion of an order must cascade to its line items.
FR-O11	Should	A promotion must capture its type (Percentage, Fixed, or BOGO), its discount value, and its validity period (start date and end date inclusive). End date must be on or after start date.

5.3 Product & Catalog Management

ID	Priority	Requirement
FR-P01	Must	Each product must have a unique system-assigned identifier and a unique SKU code.
FR-P02	Must	Product name and list price must be stored. List price is the current published selling price and may change over time (no price history is required).
FR-P03	Must	Each product must be assigned to exactly one leaf-level category (Level 3 in the hierarchy). Assignment to root or mid categories is not permitted.
FR-P04	Must	The product category hierarchy must support up to three levels: Root, Mid, and Leaf. Each category must record its level and, for non-root categories, a reference to its parent category.
FR-P05	Must	Each product must be linked to exactly one primary supplier at any point in time.
FR-P06	Must	Supplier records must capture name, country, standard lead time in calendar days, and an optional contact email.
FR-P07	Must	Products must be soft-deletable via an IsActive flag. Inactive products must be retained to preserve order history integrity.
FR-P08	Should	Cost price per unit from the supplier must be optionally recordable on the product. This field is restricted to finance roles and must be protected by column-level access control in Snowflake.
FR-P09	Should	For analytical queries, the category hierarchy must be navigable at all three levels simultaneously, supporting both leaf-level detail and root-level roll-up without requiring recursive joins.

5.4 Analytical & Reporting Requirements

ID	Priority	Requirement
FR-A01	Must	The analytical model must support calculation of net revenue per order, per customer, per product, per category, per channel, per promotion, and per date (day, week, month, quarter, year) without post-processing.

ID	Priority	Requirement
FR-A02	Must	Customer lifetime value must be queryable as a snapshot value associated with a specific point in time, supporting cohort analysis by segment.
FR-A03	Must	Promotional effectiveness must be analyzable by comparing net revenue on orders with a promotion applied against those without.
FR-A04	Must	The date dimension must support filtering and grouping by day, day of week, week, month, month name, quarter, and year. Weekend and public holiday indicators must be included.
FR-A05	Must	Channel-level revenue analysis must support grouping by individual channel (Web, Mobile, In-Store, Partner) and by channel group (Digital vs Physical).
FR-A06	Should	Gross margin per line item must be calculable for users with access to cost data, as net revenue minus the cost amount (standard cost multiplied by quantity).
FR-A07	Should	The fact table grain must be one row per order line item to support the most granular level of analysis without pre-aggregation.
FR-A08	Could	The analytical model must be extensible to support future addition of a returns domain without requiring restructuring of the core fact or dimension tables.

6 Data Governance & Quality Requirements

6.1 PII Classification

The following attributes contain personally identifiable information and must be classified accordingly in the data model. Classification must be expressed as a model-level annotation (note) and enforced as a Snowflake column masking policy in all environments other than production.

ID	Priority	Requirement
FR-G01	Must	Customer Email must be classified as PII. In non-production environments, it must be replaced with a tokenized or masked value that preserves format but contains no real data.
FR-G02	Must	Customer FirstName and LastName must be classified as PII and masked in non-production.
FR-G03	Must	Customer Address (all fields) must be classified as PII and masked in non-production.
FR-G04	Must	Product CostPrice must be classified as a restricted financial metric. Access must be limited to finance roles via Snowflake row-access or column-masking policy.
FR-G05	Must	All PII and restricted attributes must carry a classification annotation in the data model definition, not solely in external documentation.

6.2 Data Quality Rules

ID	Priority	Requirement
FR-Q01	Must	Customer Email must conform to a valid email format. Invalid formats must be rejected at the point of ingestion.
FR-Q02	Must	Order line item Quantity must be greater than zero. A zero-quantity line item must be rejected.
FR-Q03	Must	DiscountPct on a line item must be in the range 0.0000 to 1.0000 inclusive.
FR-Q04	Must	Promotion EndDate must be on or after StartDate.
FR-Q05	Must	A product must be assigned to a leaf-level category (Level = 3). Assignment to Level 1 or Level 2 categories must be rejected at the application layer.
FR-Q06	Must	Order TotalAmount must be greater than or equal to zero.
FR-Q07	Should	DiscountValue on a promotion must be greater than or equal to zero.
FR-Q08	Should	Supplier LeadTimeDays must be a positive integer greater than zero.

7 Non-Functional & Modeling Constraints

7.1 Target Platform

Platform	The physical data model must target Snowflake on Azure as the primary analytical platform. All column types, constraints, and index definitions must use Snowflake-compatible DDL syntax. The logical model may use platform-neutral data types (varchar, boolean, decimal, timestamp, date, int).

7.2 Modeling Constraints

- The logical model must be in third normal form (3NF). No denormalization is permitted at the logical layer.
- The physical model must implement a star schema pattern: a central fact table at line-item grain surrounded by conformed dimensions.
- Surrogate keys (UUID, varchar(36)) must be used throughout the physical model. Natural keys must be retained as non-key attributes for traceability.
- Slowly changing dimensions must be implemented as SCD Type 2 for the customer dimension, recording effective-from and effective-to dates and an is-current flag.
- The category hierarchy must be deformalized into three fixed-depth columns (L1, L2, L3) in the product dimension to avoid recursive join requirements in BI tools.

- All monetary measures must be stored as decimal(12,2). Fractional measures (discount percentages, ratios) must be stored as decimal(5,4).
- All timestamps in the logical model must be stored in UTC. The physical model must use Snowflake's TIMESTAMP_NTZ type.
- The data model output must be expressed in DBML format (https://dbml.dbdiagram.io/home/) and must be parseable by dbdiagram.io without errors.

7.3 Modeling Output Specification

The data modeler must produce three DBML artifacts within a single file, clearly separated by layer:

- Layer 1 - Conceptual: Entities and relationships only, with no attributes. Each entity carries a single placeholder column to satisfy DBML parser requirements. Relationships must use DBML Ref syntax with direction operators (< for one-to-many, > for many-to-one, - for one-to-one). Self-referencing entities must use two distinct column names as endpoints to avoid DBML error 5002.
- Layer 2 - Logical (3NF): All entities with full attribute definitions, platform-neutral data types, primary keys, foreign keys, unique constraints, not null constraints, and default values. Inline FK references using DBML ref syntax on column definitions. Table-level and column-level notes for business definitions and governance classification.
- Layer 3 - Physical (Snowflake Star Schema): Denormalised star schema with surrogate keys, Snowflake-specific column types, all constraints expressed as DBML column attributes, index definitions for all foreign keys and composite analytical access patterns, and governance annotations on all PII and restricted columns.

DBML-specific compliance rules that must be observed:

- Note values must be single-line quoted strings. Multi-line notes using literal newlines within quotes will cause a parse error.
- Em dashes and other non-ASCII characters must not appear in note strings or inline comments.
- Ref annotations ([note: ...]) are not valid on standalone Ref declarations; they are only valid on column definitions.
- Self-referencing relationships must use two distinct column names as endpoints; identical endpoints on both sides of a Ref trigger error 5002.
- SQL reserved words used as table names (e.g. ORDER) do not require quoting in DBML table declarations, but quoted names can cause issues in TableGroup and Ref contexts.

8 Business Glossary

The following terms are used throughout this document and must be reflected consistently in data model entity and attribute names.

Term	Definition
Customer	A person or organization registered with RetailCo or having completed at least one purchase. Identified by a unique system-assigned ID across all channels.
Segment	A tier classification assigned to a customer based on their lifetime value. Valid values: Gold, Silver, Bronze, New. Managed as a separate entity referenced by FK from Customer.
Lifetime Value	The cumulative net revenue attributed to a customer across all orders, regardless of channel. Updated on each confirmed order.
Acquisition Channel	The channel through which a customer first engaged with RetailCo. Set once at registration. Values: Web, Store, Referral, Partner.
Order	A purchase transaction placed by a customer. Has one or more line items and passes through a defined status lifecycle.
Channel	The commercial channel through which an order was placed. Values: Web, Mobile, In-Store, Partner. Grouped into Digital (Web, Mobile) and Physical (In-Store, Partner).
Line Item	A single product entry within an order, recording product, quantity, unit price, and line-level discount.
Extended Price	The revenue attributable to a single line item, calculated as Quantity x UnitPrice x (1 - DiscountPct). Additive across line items.
Gross Order Value	The sum of all line item extended prices on an order, before application of any order-level promotional discount.
Net Revenue	Gross order value minus the order-level discount amount. The primary revenue metric for business reporting.
Promotion	A discount campaign optionally applied to an order. Types: Percentage (fraction of gross value), Fixed (absolute amount), BOGO (buy-one-get-one).
List Price	The standard published selling price of a product at the current point in time. May differ from the unit price charged on a historical order.
Unit Price	The actual price charged for a product on a specific order line item. Captured at time of order and does not change if list price is later updated.
Cost Price	The per-unit cost of a product from its primary supplier. A restricted financial metric visible to finance roles only.
Standard Cost	Cost price as used in analytical margin calculations: CostPrice x Quantity = CostAmount per line item.
Gross Margin	Net revenue minus cost amount for a given line item. Calculable only when cost price is available.
SKU	Stock-Keeping Unit. A unique alphanumeric code that identifies a product across operational and analytical systems.

Term	Definition
Category	A hierarchical classification for products with up to three levels: Root, Mid, and Leaf. Products are assigned only to Leaf categories.
Supplier	A vendor that provides products to RetailCo. Each product has one primary supplier.
Lead Time	The standard number of calendar days between placing a purchase order with a supplier and receiving the goods.
SCD Type 2	Slowly Changing Dimension Type 2. A technique for tracking historical attribute changes by adding new rows with effective-from and effective-to dates, preserving a full audit trail.
Surrogate Key	A system-generated unique identifier (UUID) used as the primary key in dimensional tables, replacing the natural business key to isolate the warehouse from source system key changes.
Degenerate Dimension	A dimension attribute stored directly on the fact table rather than in a separate dimension table. OrderID is a degenerate dimension on FACT_ORDER_LINE.
PII	Personally Identifiable Information. Attributes that can identify a natural person. Requires classification, access control, and masking in non-production environments.
Product	A sellable item in the RetailCo catalogue. Assigned to exactly one leaf-level category and one primary supplier.
Order Line Item	A single product entry within an order. The atomic grain of purchase data.
Customer Segment History	History of segment assignments. Enables point-in-time cohort analysis at any historical date.

9 Assumptions & Exclusions

9.1 Assumptions

- Customer identity resolution across legacy source systems will be performed prior to data loading. The data model assumes a single, resolved CustomerID per natural person is supplied by the ingestion layer.
- Currency is assumed to be a single base currency. Multi-currency conversion is out of scope for this initiative; all monetary values are stored in the base currency.
- The holiday calendar for IS_HOLIDAY in the date dimension will be maintained by the data engineering team and scoped to the primary trading country.
- Product cost data may not always be available at the time of order line item loading. The COST_AMOUNT and GROSS_MARGIN measures on the fact table are nullable for this reason.
- Category hierarchy depth is fixed at a maximum of three levels for this initiative. Deeper hierarchies require a model revision.

- A product has exactly one primary supplier at any point in time. Supplier history (changes of primary supplier over time) is not required in this version.

9.2 Exclusions

- Returns and refunds domain: out of scope for this initiative. The physical model must be extensible to accommodate returns in a future phase.
- Inventory and stock management: out of scope. Product quantity on hand is not modeled.
- Pricing rules and price book management: out of scope. Only the current list price per product is captured.
- Customer authentication and account management: out of scope. The data model captures analytical attributes only, not authentication credentials.
- Multi-currency support: out of scope. See assumptions above.
- Supplier performance and purchase orders from RetailCo to suppliers: out of scope.